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In [15]: `!pip install "dask[dataframe]"`

```
Requirement already satisfied: dask[dataframe] in /usr/local/lib/python3.12/dist-packages (2026.3.0)
Requirement already satisfied: click>=8.1 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (8.4.1)
Requirement already satisfied: cloudpickle>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (3.1.2)
Requirement already satisfied: fsspec>=2021.09.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (2025.3.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (26.2)
Requirement already satisfied: partd>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (1.4.2)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (6.0.3)
Requirement already satisfied: toolz>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (0.12.1)
Requirement already satisfied: pandas>=2.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (2.2.2)
Requirement already satisfied: pyarrow>=16.0 in /usr/local/lib/python3.12/dist-packages (from dask[dataframe]) (18.1.0)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.0->dask[dataframe]) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.0->dask[dataframe]) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.0->dask[dataframe]) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.0->dask[dataframe]) (2026.2)
Requirement already satisfied: locket in /usr/local/lib/python3.12/dist-packages (from partd>=1.4.0->dask[dataframe]) (1.0.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=2.0->dask[dataframe]) (1.17.0)
```

In [8]: `from google.colab import files
uploaded = files.upload()`

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Sample - Superstore.csv.zip to Sample - Superstore.csv (2).zip

In [9]: `import zipfile

with zipfile.ZipFile('Sample - Superstore.csv.zip', 'r') as zip_ref:
 zip_ref.extractall()

print("File Extracted Successfully!")`

File Extracted Successfully!

In [10]: `import dask.dataframe as dd

df = dd.read_csv(
 "Sample - Superstore.csv",
 encoding="latin1"
)

print(df.head())`

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	

	Customer Name	Segment	Country	City	...	\
0	Claire Gute	Consumer	United States	Henderson	...	
1	Claire Gute	Consumer	United States	Henderson	...	
2	Darrin Van Huff	Corporate	United States	Los Angeles	...	
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	

	Postal Code	Region	Product ID	Category	Sub-Category	\
0	42420	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311	South	FUR-TA-10000577	Furniture	Tables	
4	33311	South	OFF-ST-10000760	Office Supplies	Storage	

		Product Name	Sales	Quantity	\
0		Bush Somerset Collection Bookcase	261.9600	2	
1		Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2		Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3		Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4		Eldon Fold 'N Roll Cart System	22.3680	2	

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714

```
3      0.45 -383.0310
4      0.20   2.5164
```

[5 rows x 21 columns]

```
In [11]: # Dataset information
print("Number of rows:", df.shape[0].compute())
print("Number of columns:", len(df.columns))
```

Number of rows: 9994
Number of columns: 21

```
In [12]: # Total sales by category
sales_by_category = df.groupby("Category")["Sales"].sum().compute()
print(sales_by_category)
```

Category
Furniture 741999.7953
Office Supplies 719047.0320
Technology 836154.0330
Name: Sales, dtype: float64

```
In [13]: # Total profit by region
profit_by_region = df.groupby("Region")["Profit"].sum().compute()
print(profit_by_region)
```

Region
Central 39706.3625
East 91522.7800
South 46749.4303
West 108418.4489
Name: Profit, dtype: float64

```
In [14]: # Top 10 products by sales
top_products = df.groupby("Product Name")["Sales"].sum().nlargest(10).compute()
print(top_products)
```

Product Name
Canon imageCLASS 2200 Advanced Copier 61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind 27453.384
Cisco TelePresence System EX90 Videoconferencing Unit 22638.480
HON 5400 Series Task Chairs for Big and Tall 21870.576
GBC DocuBind TL300 Electric Binding System 19823.479
GBC Ibimaster 500 Manual ProClick Binding System 19024.500
Hewlett Packard LaserJet 3310 Copier 18839.686
HP Designjet T520 Inkjet Large Format Printer - 24" Color 18374.895
GBC DocuBind P400 Electric Binding System 17965.068
High Speed Automatic Electric Letter Opener 17030.312
Name: Sales, dtype: float64